Quaternionic Rigidity of Admissible Morphisms: Every Admissible $\Phi_{q,\rho}$ Necessarily Factors through $\mathfrak{su}(2)$

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2026

Abstract

Let $\Phi_{q,\rho}\colon V_q \to V_\rho$ be a morphism from the Weil representation space of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ to a representation space of $2I \subset \mathrm{SU}(2)$, subject to three structural constraints imposed by the admissible projection Π: involution equivariance $(\chi \mapsto -\chi)$, quadratic compatibility with the pair observable σ_{pair}, and naturality with respect to admissibility. The main result of this paper is a *rigidity theorem*: any morphism satisfying these three constraints is forced to factor through the three-dimensional real Lie algebra $\mathfrak{su}(2)$, identified with the admissible neutral traceless sector $\mathrm{Im}\,\mathbb{H} = \mathrm{Im}\,\Pi \cap \mathrm{Ntrl}$. The quaternionic structure is not a choice but the unique fixed point of the three constraints. This upgrades the result of O23 [8]—which established $\dim_{\mathbb{R}}(\mathrm{Im}\,\mathbb{H}) = 3$—from a dimension count to a categorical universality statement: $\mathfrak{su}(2)$ is the unique minimal admissible target through which every $\Phi_{q,\rho}$ must factor. Combined with the chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$ of O24 [9], this gives β^* a representation-theoretic meaning as the effective scaling exponent of norm growth in the minimal admissible non-abelian sector. The canonical construction $\Phi_{q,\rho} = \rho \circ \iota \circ \pi$ is exhibited explicitly and proved unique up to unitary equivalence.

Contents

1 Introduction

1.1 The problem: existence of an admissible morphism

Paper O26 [4] identified the pair observable $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ as the Hilbert–Schmidt norm squared of a rank-one operator in $\operatorname{End}(V_\rho)$ for some representation ρ of $2I \subset \mathrm{SU}(2)$. This identification rests on the existence of a morphism

$$\Phi_{q,\rho} \colon V_q \longrightarrow V_\rho$$

satisfying three structural constraints:

(a) *Involution equivariance*: $\mathbb{Z}_2$-equivariance under $c \leftrightarrow q-c$, the discrete realisation of $\chi \mapsto -\chi$;

(b) *Quadratic compatibility*: $\sigma_{\text{pair}}(n) \sim \|\Phi_{q,\rho}(v_c^{(n)})\|^2 \cdot \|\Phi_{q,\rho}(v_{q-c}^{(n)})\|^2$;

(c) *Naturality*: independence from all pipeline choices; dependence on the admissible structure $(\Pi, S_{\mathrm{BI}}, Q_8)$ alone.

Paper O26 stated the existence of such a $\Phi_{q,\rho}$ as Hypothesis 4.4. The present paper resolves this hypothesis. The resolution is stronger than mere existence: the three constraints force a unique factorisation, making $\Phi_{q,\rho}$ a *structural necessity* rather than a canonical choice.

1.2 The answer: a rigidity theorem

The central result is the following. Any morphism $\Phi_{q,\rho}$ satisfying (a)–(c) must factor through the admissible quotient $H_{\text{eff}} = \operatorname{Im}\Pi \cap \mathrm{Ntrl}$, and the quaternionic identification $H_{\text{eff}} \simeq \mathfrak{su}(2)$ (O23 [8]) is forced by the interplay of the three constraints. No other three-dimensional or higher-dimensional real structure is compatible with all three constraints simultaneously. The comparison is explicit (Table 1):

1.3 Relation to the O-series chain

The transfer $c_\chi \to \delta_{\text{pair}} \to \beta^*$ was proved unconditionally in O24 [9]. The present result adds a representation-theoretic layer to that chain:

emergence $\Rightarrow$ non-injectivity $\Rightarrow$ pair structure $\Rightarrow$ quadratic form $\Rightarrow$ $\mathfrak{su}(2)$,

Structure	(a)	(b)	(c)	Verdict
Abelian $\mathbb{R}^n$	$\times$	$\times$	✓	excluded by O23
Standard $\mathbb{C}^n$	✓	✓	$\times$	not canonical
Arbitrary Hilbert space	✓	✓	$\times$	pipeline-dependent
$\mathbb{H}\,/\,\mathfrak{su}(2)$	✓	✓	✓	unique candidate

Table 1: Compatibility of candidate target structures with constraints (a)–(c).

where each implication is a theorem of the O-series (the first being the no-go theorem of paper L [3]). The exponent β^* thereby acquires a direct representation-theoretic meaning: it is the effective scaling exponent of norm growth along a trajectory in the minimal admissible non-abelian sector $\mathfrak{su}(2)$.

1.4 Scope

This paper proves the rigidity theorem (Section 4) and the canonical construction (Section 5). It does not execute the numerical tests of O26 (those require O25 data [7]). It does not revisit the transfer chain, proved in O24 [9].

1.5 Organisation

Section 2 fixes notation and recalls the relevant results from O18–O24. Section 3 eliminates the abelian and generic Hilbert-space candidates. Section 4 states and proves the rigidity theorem. Section 5 exhibits the canonical factorisation and proves uniqueness. Section 6 derives the representation-theoretic interpretation of β^*. Section 7 states the remaining open problem.

2 Setup and Notation

Let q be an odd prime. The Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ on $\mathbb{C}^q$ decomposes over conjugate character pairs $\{c, q-c\}$ with $c \in (\mathbb{Z}/q\mathbb{Z})^\times$, yielding blocks $V_c \simeq \mathbb{C}^q$ and $V_{q-c} \simeq \overline{V_c}$. The pair space is $V_q := V_c \oplus V_{q-c}$.

The parity involution $\chi \mapsto -\chi$, derived in O18 [2] from the Born–Infeld evenness of the substrate action, acts on V_q by complex conjugation: $V_{q-c} \simeq \overline{V_c}$. The canonical pair observable (O19 [1]) is

$$\sigma_{\mathrm{pair}}(n) := \sigma_c(n) \cdot \sigma_{q-c}(n), \qquad \sigma_c(n) = \delta r_n/|S_n|.$$

The admissible neutral traceless sector is

$$H_{\mathrm{eff}} := \mathrm{Im}\,\Pi \cap \mathrm{Ntrl},$$

with $\dim_{\mathbb{R}} H_{\mathrm{eff}} = 3$ by O23 Theorem 3.2 [8], together with the quaternionic identification

$$\iota\colon H_{\mathrm{eff}} \xrightarrow{\simeq} \mathfrak{su}(2),$$

canonical and unique up to SU(2)-conjugacy. The admissibility thread $Q_8 \subset 2I \subset \mathrm{SU}(2)$ selects a spin-$\frac{1}{2}$ representation ρ as the minimal non-abelian candidate (O26 [4], Section 3.3).

Definition 2.1 (Naturality with respect to admissibility)**.** *A morphism* $\Phi\colon V_q \to W$ *is* natural with respect to admissibility *if it is invariant under all admissible endomorphisms of* V_q*, i.e. under every linear map* $f\colon V_q \to V_q$ *that preserves* Π*-admissibility in the sense that* $f(\ker \Pi_{\mathrm{adm}}) \subseteq \ker \Pi_{\mathrm{adm}}$*. Equivalently,* Φ *is natural if it depends only on the admissible structure* $(\Pi, S_{\mathrm{BI}}, Q_8)$ *and not on any auxiliary choice of basis, BFS block, or Gram–Schmidt frame.*

Throughout, V_ρ denotes a complex representation space of $2I$ of dimension d_ρ, endowed with its natural Hilbert structure.

Remark 2.2 (Non-arbitrariness of ρ via the admissibility thread)*.* The representation ρ is not chosen freely. The admissibility thread

$$Q_8 \;\subset\; 2I \;\subset\; \mathrm{SU}(2)$$

is imposed structurally: Q_8 is the minimal non-abelian finite subgroup of SU(2) compatible with bounded-flux admissibility [5]; $2I$ is the maximal admissible finite subgroup, distinguished by the isotropy of its second-moment tensor among all binary polyhedral groups [6]; and the LPS family provides the Ramanujan approximation to SU(2) in the large-q limit [5]. The spin-$\frac{1}{2}$ sector is selected because $Q_8 \subset 2I$ acts on $V_{\rho_{1/2}} \simeq \mathbb{C}^2$ as the minimal non-abelian representation, yielding the smallest non-trivial Laplacian eigenvalue $\lambda_{\rho_4} = 6 < 8 = \lambda_{\mathrm{ab}}$ and hence the widest admissibility window [5, 4]. The choice of ρ is therefore an output of the admissibility hierarchy, not an input.

3 Elimination of Non-Quaternionic Candidates

We show that no candidate target structure other than $\mathfrak{su}(2)$ is compatible with all three constraints.

Proposition 3.1 (Abelian structures are excluded)**.** *No abelian target* V_ρ *satisfies conditions (a), (b), and (c) simultaneously.*

Proof. Condition (c) forces $\Phi_{q,\rho}$ to factor through $H_{\mathrm{eff}} = \mathrm{Im}\,\Pi \cap \mathrm{Ntrl}$. By O23 Theorem 3.2 [8], H_{eff} is quaternionic and its real rank is exactly 3, maximal among all admissible associative structures compatible with the Born–Infeld parity. An abelian target would require H_{eff} to split as a direct sum of one-dimensional real factors, contradicting the non-commutativity of the quaternionic structure. $\square$

Proposition 3.2 (Generic complex Hilbert spaces are not canonical)**.** *A morphism* $\Phi_{q,\rho}$ *targeting an arbitrary complex Hilbert space* $\mathcal{H}$ *can satisfy* *conditions (a) and (b) but cannot satisfy condition (c) unless $\mathcal{H}$ admits a quaternionic structure compatible with admissibility.*

Proof. Conditions (a) and (b) can be satisfied by any space admitting a conjugation and a norm, hence in particular by any $\mathcal{H}$. Condition (c) requires that $\Phi_{q,\rho}$ depend only on $(\Pi, S_{\mathrm{BI}}, Q_8)$. Any such morphism must factor through the admissible quotient $\pi\colon V_q \to H_{\mathrm{eff}}$ (universality of the admissible quotient; see Lemma 3.3). The image H_{eff} is three-dimensional and quaternionic. A morphism $H_{\mathrm{eff}} \to \mathcal{H}$ that is natural with respect to admissibility must intertwine the SU(2)-action, hence it is a morphism of representations. By Schur's lemma, such a morphism exists and is canonical only when $\mathcal{H}$ contains V_ρ as an SU(2)-subrepresentation, i.e. when $\mathcal{H}$ carries a compatible quaternionic structure. □

Lemma 3.3 (Universality of the admissible quotient)**.** *Let $\pi\colon V_q \twoheadrightarrow H_{\mathrm{eff}} = V_q/\ker \Pi_{\mathrm{adm}}$ be the admissible quotient map. Any morphism $\Phi\colon V_q \to W$ satisfying condition (c) factors uniquely through π: there exists $\bar{\Phi}\colon H_{\mathrm{eff}} \to W$ such that $\Phi = \bar{\Phi} \circ \pi$.*

Proof. Condition (c) states that Φ depends only on the admissible structure. The kernel $\ker \Pi_{\mathrm{adm}}$ consists precisely of the directions invisible to admissible observables. Any morphism that is natural with respect to admissibility must vanish on $\ker \Pi_{\mathrm{adm}}$, for otherwise it would distinguish substrate configurations that are admissibility-equivalent. Vanishing on $\ker \Pi_{\mathrm{adm}}$ is exactly the factorisation condition through π. □

4 Rigidity Theorem

Theorem 4.1 (Quaternionic rigidity)**.** *Let $\Phi_{q,\rho}\colon V_q \to V_\rho$ be any morphism satisfying conditions (a)–(c). Then:*

(i) *$\Phi_{q,\rho}$ factors through the admissible quotient $\pi\colon V_q \to H_{\mathrm{eff}}$;*

(ii) *the intermediate space H_{eff} is necessarily isomorphic to $\mathfrak{su}(2)$ as a real Lie algebra, via the quaternionic identification ι of O23 [8];*

(iii) *the target V_ρ is necessarily an SU(2)-representation space, and the morphism $H_{\mathrm{eff}} \to V_\rho$ is a representation morphism.*

In particular, under constraints (a)–(c), any admissible morphism $\Phi_{q,\rho}$ necessarily factors through a three-dimensional real Lie algebra isomorphic to $\mathfrak{su}(2)$. The quaternionic structure is not a modelling choice but the unique fixed point of the three constraints. In particular, the result is independent of the specific Weil realisation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ *and holds for any admissible representation of the pair structure satisfying (a)–(c), whatever the underlying group or prime q.*

Proof. **Part (i).** Follows immediately from Lemma 3.3: condition (c) forces factorisation through π.

Part (ii). After factorisation, $\bar{\Phi}\colon H_{\mathrm{eff}} \to V_\rho$ must be natural with respect to the Born–Infeld admissibility. The only admissible automorphisms of H_{eff} are those compatible with the Born–Infeld parity (condition (a)) and the quadratic norm (condition (b)). By O23 Theorem 3.2 [8], the group of such automorphisms is SU(2), acting on $H_{\mathrm{eff}} \simeq \mathfrak{su}(2)$ by the adjoint action. No other Lie group structure on a three-dimensional real space is compatible with both the conjugation involution of (a) and the positive-definite quadratic form of (b) simultaneously; this is the classical uniqueness of $\mathbb{H}$ as the only normed associative real division algebra of dimension greater than 2 (Frobenius theorem). Hence $H_{\mathrm{eff}} \simeq \mathfrak{su}(2)$ is forced.

Part (iii). A natural morphism $\bar{\Phi}\colon \mathfrak{su}(2) \to V_\rho$ must intertwine the SU(2)-action. By Schur's lemma, such a morphism exists and is non-zero only when V_ρ contains an irreducible SU(2)-subrepresentation; it is then a morphism of representations. $\square$

Remark 4.2 (Relation to O23). O23 [8] established that $\dim_{\mathbb{R}} H_{\mathrm{eff}} = 3$. Theorem 4.1 upgrades this from a dimension statement to a *categorical universality* statement: $\mathfrak{su}(2)$ is the unique minimal admissible target through which every morphism satisfying (a)–(c) must factor. The integer 3 is not an accident of the Weil representation: it is the unique dimension compatible with the simultaneous constraints of involution, quadratic form, and associativity required by naturality.

Remark 4.3 (Frobenius and admissibility). The appeal to the Frobenius theorem in Part (ii) is not circular. Frobenius establishes that $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ are the only normed real associative division algebras. Dimension 1 ($\mathbb{R}$) is excluded by the non-commutativity imposed by (c) and O23. Dimension 2 ($\mathbb{C}$) is excluded because a complex structure on H_{eff} is not preserved by the $\mathbb{Z}_2$-involution of (a) (complex conjugation is not an inner automorphism of $\mathbb{C}$). Dimension 4 ($\mathbb{H}$ itself, not $\mathfrak{su}(2) \simeq \operatorname{Im}\mathbb{H}$) is excluded because $\dim_{\mathbb{R}} H_{\mathrm{eff}} = 3$ (O23). The unique compatible structure is therefore $\operatorname{Im}\mathbb{H} \simeq \mathfrak{su}(2)$. The argument uses only the algebraic consequences of constraints (a)–(c) and the dimension bound of O23; it does not depend on the Heisenberg group structure or on properties specific to any prime q.

5 Canonical Construction and Uniqueness

Definition 5.1 (Canonical admissible morphism). *Define*

$$\Phi_{q,\rho} \;:=\; \rho \circ \iota \circ \pi,$$

where $\pi\colon V_q \to H_{\mathrm{eff}}$ *is the admissible quotient map,* $\iota\colon H_{\mathrm{eff}} \xrightarrow{\simeq} \mathfrak{su}(2)$ *is the canonical quaternionic identification of O23, and* $\rho\colon \mathfrak{su}(2) \to \operatorname{End}(V_\rho)$ *is the Lie algebra action of the chosen irreducible* SU(2)*-representation.*

Theorem 5.2 (Existence, conditions, and uniqueness)**.** *The morphism $\Phi_{q,\rho}$ of Definition 5.1 satisfies conditions (a)–(c). It is the unique morphism doing so, up to unitary equivalence: any other such morphism is of the form $U \circ \Phi_{q,\rho}$ for some unitary $U \in \mathrm{U}(V_\rho)$ commuting with the Q_8-action.*

Proof. **Conditions.** (a) The involution $\chi \mapsto -\chi$ acts on $\mathfrak{su}(2)$ by $X \mapsto -X$, hence on V_ρ by $\rho(-X) = -\rho(X)$; the $\mathbb{Z}_2$-equivariance is preserved at each stage. (b) The norm $\|\Phi_{q,\rho}(v)\|^2 = \|\rho(\iota(\pi(v)))\|^2_{V_\rho}$ recovers the quadratic structure of σ_{pair} since π preserves the norm class, ι is an isometry up to a fixed scale, and ρ is unitary. (c) Each factor π, ι, ρ depends only on $(\Pi, S_{\text{BI}}, Q_8)$, by Lemma 3.3, O23, and the admissibility thread respectively.

Uniqueness. By Theorem 4.1, any $\Phi'_{q,\rho}$ satisfying (a)–(c) factors as $\rho' \circ \iota \circ \pi$ for some representation morphism ρ'. The ambiguity in ι is SU(2)-conjugacy, which is unobservable in $\|\Phi_{q,\rho}(\cdot)\|^2$. The ambiguity in ρ' versus ρ is, by Schur's lemma, a unitary U commuting with the Q_8-action. □

6 Representation-Theoretic Interpretation of β^*

Under Theorem 5.2, the pair observable satisfies

$$\sigma_{\text{pair}}(n) \;\sim\; \left\|\Phi_{q,\rho}(v_c^{(n)}) \otimes \Phi_{q,\rho}(v_{q-c}^{(n)})\right\|^2_{\text{HS}},$$

realising δ_{pair} as the growth exponent of a Hilbert–Schmidt norm along a trajectory $(\widetilde{M}^{(n)})_{n\geq 0}$ in $\mathrm{End}(V_\rho)$.

Corollary 6.1 (Representation-theoretic meaning of β^*)**.** *Under the unconditional transfer $c_\chi \to \delta_{\text{pair}} \to \beta^*$ of O24 [9], the exponent β^* is the effective scaling exponent of norm growth in the minimal admissible non-abelian sector $\mathfrak{su}(2)$. The phenomenological window $\beta^* \in (0.09, 0.13)$ is a constraint on admissible trajectories in the representation space $\mathrm{End}(V_\rho)$, not a purely numerical fit.*

Remark 6.2 (Completeness of the chain). The full implication chain is now:

emergence $\Rightarrow$ non-injectivity [3] $\Rightarrow$ pair structure [2] $\Rightarrow$ quadratic form [4] $\Rightarrow$ $\mathfrak{su}(2)$ (Theorem 4.1

Each arrow is a proved result of the O-series or companion paper programme. No arrow is an assumption.

7 Remaining Open Problem

Theorem 4.1 establishes the structure of $\Phi_{q,\rho}$ unconditionally. The single remaining open question in the representation-theoretic component of the O-series is the numerical identification of the correct irreducible sector: which value of d_ρ is realised by O25 data?

The spin-$\frac{1}{2}$ candidate ($d_\rho = 2$) predicts effective dimension $r_{\text{eff}} = d_\rho^2 = 4$ in the covariance test of O26 Criterion 5.4 [4]. A different fixed value $r_{\text{eff}} = d_\rho^2 \neq 4$ would identify an alternative irreducible sector without invalidating the rigidity theorem, since Theorem 4.1 holds for any ρ. This test is executable on existing O25 data [7] without additional BFS computation.

Acknowledgements

The author acknowledges the use of large language models as a supportive tool for refining language, structure, and internal consistency during the development of this manuscript.

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